17

Lecture 17

3GPP Standardisation

استانداردسازی 3GPP

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part III – Software-driven architectures (Lectures 15–18)

ESTIMATED STUDY TIME
5–7 hours (3 h reading, 2–4 h exercises)

DIFFICULTY
Introductory to intermediate

PREREQUISITES
Lectures 1–16 (as examples); basic probability

In this lecture

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-17-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** how 3GPP is organised: organisational partners, TSGs and working groups, and its relation to ITU-R and the IETF.
- **LO2** **Explain** the three-stage methodology, specification numbering and the release model.
- **LO3** **Decode** a specification version number and **classify** change requests.
- **LO4** **Trace** how a feature moves from a study item to a specification and to products.
- **LO5** **Locate and read** a specification efficiently: scope, normative text, annexes.
- **LO6** **Reason quantitatively** about interoperability, work backlogs and time to market.

2 Prerequisite knowledge

This lecture is about the process behind everything in this course. The earlier lectures provide the examples: the specifications cited there (TS 23.401, TS 29.272, TS 38.300, …) are the products of the process described here. Exercise 5 uses Little's law (Lecture 15).

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
3GPP	پروژهٔ مشارکت نسل سوم	Partnership of standards organisations that specifies mobile systems.
Organisational partner	شریک سازمانی	Regional/national standards body that is a member of 3GPP.
TSG / WG	گروه مشخصات فنی / گروه کاری	Decision-making groups and their technical sub-groups.
TS / TR	مشخصات فنی / گزارش فنی	Normative specification / informative study report.
Stage 1, 2, 3	مراحل ۱، ۲، ۳	Requirements, architecture, detailed protocols.
Release	انتشار	Consistent set of features and specifications frozen together.
Freeze	انجماد	Point after which only corrections are accepted.
Change request (CR)	درخواست تغییر	Formal proposal to modify a specification under change control.
Work item / study item	مورد کاری / مطالعه	Approved task to specify / to study a feature.
SEP / FRAND	پتنت ضروری استاندارد	Standard-essential patent / fair, reasonable and non-discriminatory licensing.

3 Introduction and motivation

A phone bought in one country works on a network built from equipment of several vendors in another country. That this works at all is the result of standards: thousands of documents that specify every message, every timer and every bit, written by competitors who cooperate in the same rooms. For mobile networks the dominant body is the *3rd Generation Partnership Project* (3GPP), created in 1998 to specify the third generation and since responsible for GSM's evolution, LTE, 5G and, now, 6G.

Engineers in the mobile industry spend much of their time with 3GPP specifications: implementing them, testing against them, or contributing to them. Understanding how the documents are organised, how they evolve and how decisions are made is therefore a practical skill, not trivia. It also explains why the systems look the way they do: many design choices in this course are compromises reached in standardisation meetings.

4 Fundamental concepts

4.1 Organisation

3GPP is not a legal entity of its own: it is a partnership of seven *organisational partners* – ARIB and TTC (Japan), ATIS (USA), CCSA (China), ETSI (Europe), TSDSI (India) and TTA (Korea) – which transpose its specifications into their own standards. Companies participate as members of one of these partners; *market representation partners* (industry associations) contribute requirements. The technical work is done in three *Technical Specification Groups* and their working groups (Figure 1) [7]:

- **TSG RAN**: radio access (physical layer, radio protocols, RAN architecture, RF, UE conformance tests);
- **TSG SA**: service and system aspects (requirements, architecture, security, codecs, management) and overall coordination;
- **TSG CT**: core network and terminals (NAS, core protocols, policy, USIM).

Working groups prepare documents; TSG plenaries, meeting quarterly, approve them. Decisions are made by consensus; formal voting is possible but rare.

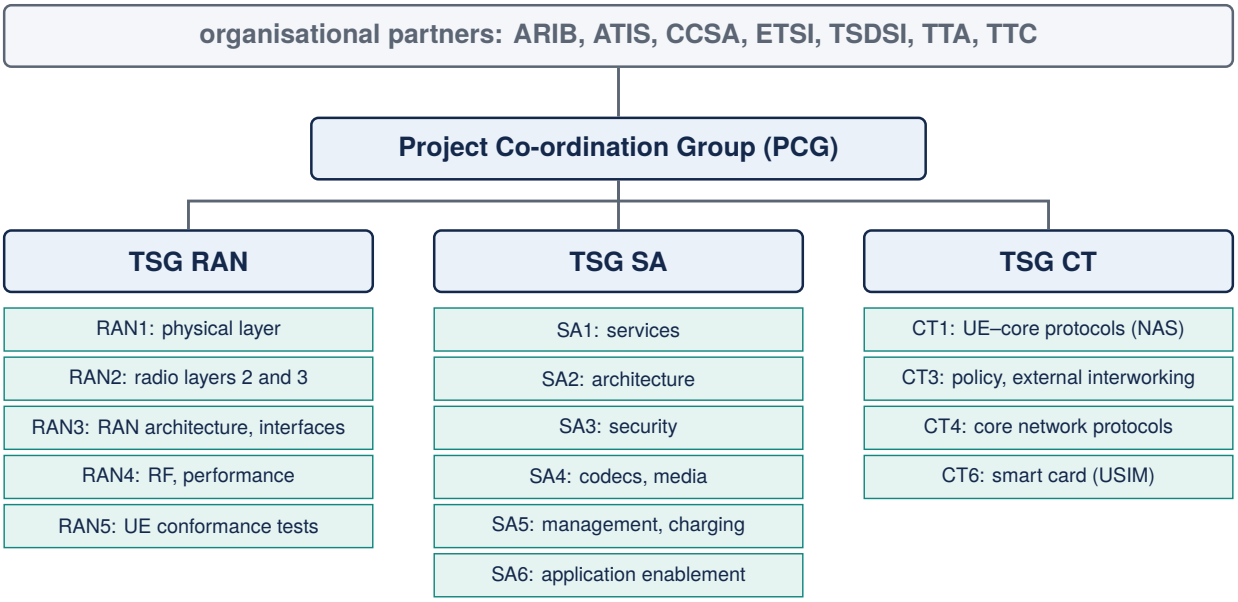


Figure 1. Organisation of 3GPP: organisational partners, project co-ordination, TSGs and their working groups (selection).

4.2 Relation to other bodies

ITU-R defines the requirements of each IMT generation (Lecture 14) [12], [13]; 3GPP's technologies are submitted, through the organisational partners, as candidate IMT radio interfaces and evaluated. The IETF specifies the Internet protocols that 3GPP reuses – SCTP, Diameter, HTTP/2, SIP, IPsec, TLS [10], [14]–[16] – and 3GPP requests extensions when needed. Industry bodies such as the GSMA define operational guidelines (e.g. roaming, VoLTE profiles), and certification organisations test devices against 3GPP conformance specifications.

4.3 Three stages

Each feature is specified at three levels (Figure 2):

- **Stage 1** describes the service from the user's point of view: requirements, without solutions (e.g. TS 22.261 [3]).
- **Stage 2** describes the architecture: functional entities, reference points and information flows (e.g. TS 23.501 [6]).
- **Stage 3** specifies protocols bit by bit: messages, information elements, encodings, state machines (e.g. TS 24.501 [1], TS 38.331 [2]).

Conformance-test specifications then define how devices are tested against stage 3.

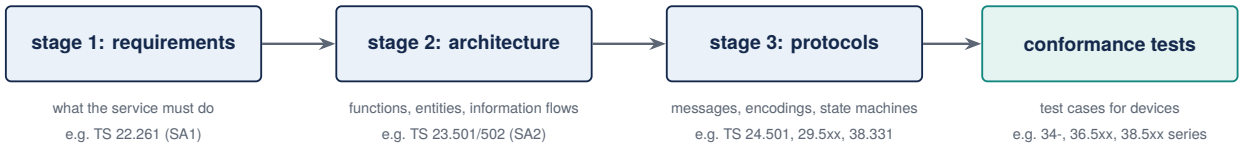


Figure 2. Three-stage methodology with typical specifications.

4.4 Specification numbering

Every specification has a number *ss.nnn* whose series *ss* indicates the subject [7]. A *technical specification* (TS) is normative: products must comply with it. A *technical report* (TR) is informative: it records studies, evaluations and options, often in the 8xx and 9xx ranges (e.g. TR 38.913 [5], TR 38.901 [4]).

Table 1. Selected 3GPP specification series.

Series	Subject	Example
21	Requirements, vocabulary, working methods	TR 21.905 [8]
22	Service aspects (stage 1)	TS 22.261
23	Technical realisation (stage 2)	TS 23.401, 23.501
24	Signalling protocols UE–core (stage 3)	TS 24.301, 24.501
26	Codecs	–
28, 32	Management and charging	TS 28.530
29	Core network protocols (stage 3)	TS 29.274, 29.500
31	SIM/USIM	–
33	Security	TS 33.401, 33.501
35	Security algorithms	TS 35.206
36	LTE (E-UTRA) radio	TS 36.300, 36.331
37	Multiple radio technologies	TS 37.340
38	NR radio	TS 38.300, 38.331

5 Detailed technical explanation

5.1 Releases

3GPP publishes its work in *releases*: consistent sets of features specified together (Figure 3). Each release has freeze dates: after the *stage 3 freeze* no new functions are added, only corrections; after the *ASN.1 freeze* the encoding of radio protocol messages is stable so that implementations become interoperable. Releases overlap: while one is being corrected, the next is under development and the one after is being studied. Vendors and operators implement a subset of features of a release, and devices declare their capabilities, so "supporting Release 17" means little without a feature list.

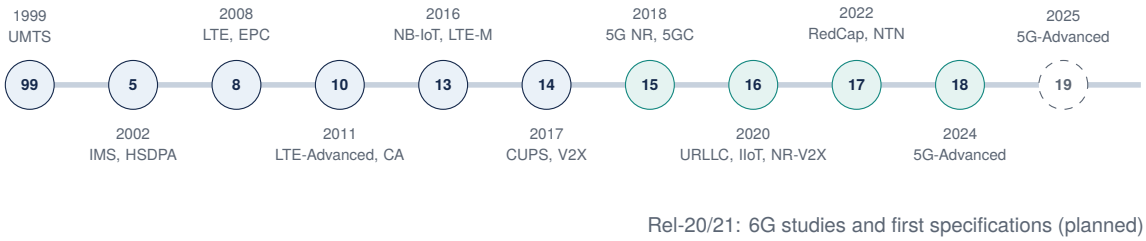


Figure 3. Selected 3GPP releases with headline features. Years indicate approximate completion; teal marks 5G releases.

5.2 Versions

Each specification has a version *x.y.z* [7]. Before approval, $x = 1$ (presented for information) or $x = 2$ (presented for approval). Once approved and under change control, x equals the release (e.g. 17 for Release 17); y increases with every set of approved technical changes; z increases for purely editorial changes. A specification thus exists in parallel versions, one per release, and corrections are often mirrored into several releases.

5.3 Change requests

A specification under change control changes only through *change requests* (CRs), each with a category [7]: **F** (correction of an error), **A** (mirror of a category-F correction in a later release), **B** (addition of a feature), **C** (functional modification of a feature) and **D** (editorial). After freeze only F, A and D are normally accepted. CRs are agreed in the working group and approved by the TSG; the specification is then republished with an incremented version (Figure 4).

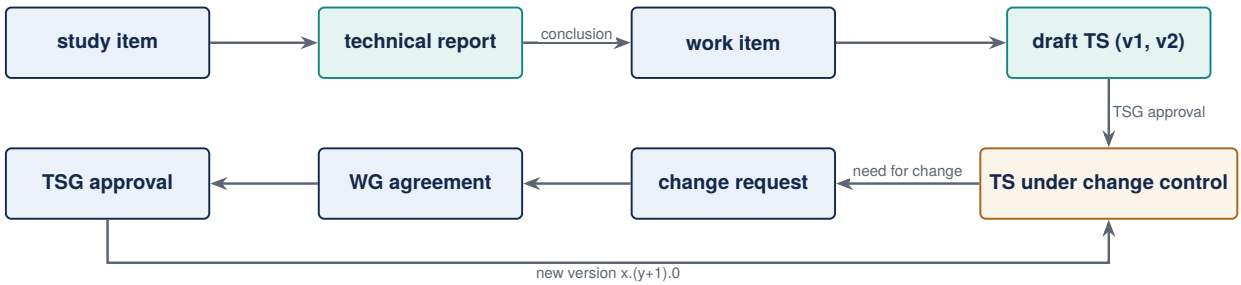


Figure 4. From study item to specification, and the change-request cycle.

5.4 How work is done

Companies submit written contributions (temporary documents, "Tdocs") to working-group meetings, held several times per release cycle in different regions, with electronic meetings in between. Proposals are discussed, merged and revised until consensus emerges; chairs record agreements. Typical contributions are proposals for solutions in a study, text proposals for a TR or TS, and CRs. All documents are public on the 3GPP website, which makes the history of every design decision traceable.

5.5 Reading a specification

A 3GPP specification follows a fixed template: scope, references, definitions and abbreviations, then the technical clauses and annexes. Normative text uses "shall" (mandatory), "should" (recommended) and "may" (permitted); informative annexes give background and examples. Practical advice: read the stage-2 description first, then the stage-3 procedure, and only then the message and information-element definitions; check the release and version; and search across specifications, because a procedure is usually split over several of them (e.g. an attach involves TS 23.401, 24.301, 36.413, 29.274 and 29.272).

5.6 Intellectual property

Companies that contribute technology often hold patents essential to implementing a standard (*standard-essential patents*, SEPs). Under the IPR policies of the organisational partners, they declare such patents and commit to license them on fair, reasonable and non-discriminatory (FRAND) terms. Licensing is negotiated bilaterally, outside 3GPP, and the level of aggregate royalties is a recurring debate (Section 6.4).

6 Mathematical models

The models below are deliberately simple; their numerical examples use illustrative assumptions, not measured 3GPP statistics.

6.1 Interoperability across interfaces

Model. An end-to-end service crosses k multi-vendor interfaces. If each is implemented correctly with probability p, independently, the service works with probability

$$P_{\text{e2e}} = p^k. \tag{1}$$

Example. $p = 0.98$, $k = 20$: $P_{\text{e2e}} = 0.67$. To reach 0.9 with $k = 20$ requires $p \geq 0.9^{1/20} = 0.995$.

Interpretation. The more interfaces a system has, the more precise the specifications and the more thorough the testing must be – the reason for stage-3 detail, conformance tests and interoperability test events. It is also an argument for fewer, simpler interfaces.

6.2 Work in progress

Model. If CRs on a specification group arrive at rate λ (per month) and each spends on average T months from first submission to approval, Little's law gives the number of open CRs:

$$N = \lambda\, T. \tag{2}$$

Example. With $\lambda = 400$ CRs/month and $T = 3$ months, about 1200 CRs are open at any time. Halving T (e.g. with electronic meetings between plenaries) halves the backlog for the same throughput.

6.3 Time to market

Model. A feature passes through study, normative work, implementation and testing; phases can partly overlap. The time from study start to commercial availability is

$$T_{\text{market}} = \sum_{i} T_i - \sum_{j} \Delta_j, \tag{3}$$

with phase durations T_i and overlaps Δ_j.

Example. Study 12 months, stage 2 12, stage 3 12, implementation 18, testing and certification 6, with 6 months of overlap between implementation and late stage 3: $T_{\text{market}} = 54$ months, i.e. about 4.5 years – consistent with the gap of several years between the start of a generation's studies and its first commercial networks.

6.4 Royalty stacking

Model. A device price P carries royalties to N SEP holders. If each independently asks a rate r_i, the total is $R = \sum_i r_i$; under an aggregate cap $R_{\max}$ allocated by each holder's share s_i of essential patents,

$$R = \sum_{i=1}^{N} r_i, \qquad r_i^{\text{cap}} = R_{\max}\, s_i, \quad \sum_i s_i = 1. \tag{4}$$

Example. Twenty holders each asking 1 % stack to 20 % of the device price; with a 5 % cap, a holder with 10 % of the essential patents receives 0.5 %. The economics of SEPs are contested; the model only shows why the question arises.

7 Network architecture and diagrams

The organisation of 3GPP mirrors the architecture of the systems it specifies: RAN working groups own the radio interface and RAN interfaces (Uu, X2/Xn, S1/NG, F1), SA2 owns the system architecture and reference points, CT groups own the core protocols and the UE–core NAS. A new feature that crosses these boundaries – network slicing, for example – needs coordinated work items in several groups, with SA2's stage 2 as the common reference.

8 Step-by-step procedures

Procedure 1 • Introducing a new feature

1. Companies propose a study item (SI) with objectives; the TSG approves it.
2. The working groups study use cases and candidate solutions and document them in a TR, with evaluations and a conclusion.
3. A work item (WI) is approved with a scope, target release and responsible groups.
4. Stage 2 and stage 3 specifications are written or extended (new TS or CRs to existing ones) until the stage 3 and ASN.1 freezes.
5. After freeze, corrections continue through category-F CRs; conformance tests are specified; products implement and certify the feature.

Procedure 2 • Finding the right specification

1. Identify the topic and stage: requirement (22), architecture (23), protocol (24, 29, 36, 38), security (33).
2. Start from the stage-2 overview (e.g. TS 23.501 or TS 38.300) and follow its references.
3. Choose the release that matches the implementation and download the latest version of that release.
4. Read scope and definitions, then the procedure clause, then the message definitions; check the CR history for recent changes.

9 Worked examples

Worked example 1 • Decoding a version

Problem. What do "TS 38.331 V17.5.0" and "TS 23.501 V2.0.0" tell you?
Solution. TS 38.331 is in series 38 (NR radio): the NR RRC protocol. V17.5.0: Release 17, under change control, fifth set of technical changes after its first Release-17 version, no editorial-only update since. TS 23.501 V2.0.0: a draft presented to the TSG for approval (before it became V15.0.0 in Release 15).

Worked example 2 • Interoperability budget

Problem. A VoLTE call involves about 20 interfaces between UE, eNB, MME, gateways, PCRF and IMS nodes from different vendors. What per-interface correctness is needed for 99 % end-to-end success?
Solution. $p \geq 0.99^{1/20} = 0.9995$. Field experience with early VoLTE, which needed extensive interoperability testing and an industry profile to narrow the options, illustrates the point.

Worked example 3 • Classifying CRs

Problem. Classify: (a) a wrong timer value in Release 16, fixed in Release 16; (b) the same fix copied into Release 17; (c) a new optional information element enabling a feature; (d) a

typo.
Solution. (a) F; (b) A; (c) B (only before the release's freeze); (d) D.

10 Technology comparison

Table 2. Standards bodies relevant to mobile networks.

Body	Role	Typical output
ITU-R	Spectrum regulation and IMT requirements	Recommendations and reports (e.g. M.2083, M.2410, M.2160)
3GPP	Specification of mobile systems	TS and TR (series 21–38)
IETF	Internet protocols	RFCs (e.g. SCTP, Diameter, HTTP/2)
ETSI	European standards; NFV, MEC industry groups	ENs, group specifications (e.g. NFV)
ONF, O-RAN Alliance	Open interfaces for SDN and RAN	Specifications (e.g. OpenFlow, O-RAN interfaces)
GSMA	Operator association	Permanent reference documents, profiles

Consensus versus speed. Open, consensus-based standardisation produces specifications that many companies can implement and that last for decades, at the cost of time and complexity: options are added to satisfy many parties. Industry alliances and open-source projects move faster but with less formal commitment; modern networks combine all of them.

11 Key takeaways

Key takeaways

- 3GPP = seven organisational partners; work in TSG RAN, SA and CT and their WGs; consensus decisions.
- Stage 1 (requirements), stage 2 (architecture), stage 3 (protocols); TS normative, TR informative.
- Series numbers identify the topic (23 architecture, 24/29 protocols, 33 security, 36 LTE, 38 NR).
- Releases freeze in stages; versions $x.y.z$ = release, technical changes, editorial changes.
- Change requests (F, A, B, C, D) are the only way to change a specification under change control.
- Interoperability decays as p^k; specifications and testing must be precise.

12 Exercises

Exercise 1 • Level 1 – conceptual • Which series?

In which series would you look for: (a) the NAS protocol of 5G; (b) the 5G security architecture; (c) the LTE MAC protocol; (d) the service requirements of 5G; (e) the GTP-U protocol?

Exercise 2 • Level 1 – conceptual • TS or TR?

Is a channel model used to evaluate candidate technologies normative? Where is it published, and why does it matter for comparing simulation results?

Exercise 3 • Level 2 – numerical • Versions

A vendor's documentation states compliance with "TS 24.501 V16.5.0". The operator's network implements V16.9.0. What kinds of differences can exist between the two, and how would you find them?

Exercise 4 • Level 2 – numerical • Interfaces and correctness

Using Eq. (1), compute P_{e2e} for $p = 0.995$ and $k = 20$ and 50. How many interfaces can a service cross at $p = 0.999$ if $P_{\text{e2e}} \geq 0.95$ is required?

Exercise 5 • Level 3 – analytical • Backlog

A working group handles 250 CRs per month with an average time to approval of 4 months. A new release adds a burst so that arrivals rise to 350 per month while approval capacity stays at 250 per month. Using Eq. (2) and a simple fluid argument, estimate the backlog after 6 months of the burst. What can the group do?

Exercise 6 • Level 3 – analytical • Overlapping phases

With the phase durations of Section 6.3, how much earlier could commercial products appear if implementation started when stage 2 is complete (overlapping stage 3 fully)? What risk does this create, and how do the freeze dates mitigate it?

Solutions

Solution to Exercise 1

(a) 24 (TS 24.501); (b) 33 (TS 33.501); (c) 36 (TS 36.321); (d) 22 (TS 22.261); (e) 29 (TS 29.281).

Solution to Exercise 2

No: channel models are informative; they are published in TRs (e.g. TR 38.901). They matter because simulation results from different companies are comparable only if they use the same channel model and assumptions, which the TR fixes.

Solution to Exercise 8

Both are Release 16; V16.9.0 includes four more sets of technical changes – mostly corrections (category F), possibly clarifications that change behaviour in corner cases. Compare the CR lists between the versions (the change history annex and the CR database) and check whether any affect implemented procedures.

Solution to Exercise 9

$0.995^{20} = 0.905$; $0.995^{50} = 0.778$. With $p = 0.999$: $k \leq \ln 0.95 / \ln 0.999 = 51.3$, i.e. at most 51 interfaces.

Solution to Exercise 10

Initially $N = 250 \times 4 = 1000$ open CRs. During the burst the backlog grows by $350 - 250 = 100$ per month: after 6 months about 1600, and by Little's law the time to approval rises to about $1600/250 = 6.4$ months. Options: more meeting time (additional electronic sessions), prioritisation of essential CRs, postponing optional features to the next release.

Solution to Exercise 11

Implementation would overlap the whole 12 months of stage 3 instead of 6: products about 6 months earlier (48 instead of 54 months). Risk: implementing a moving target – later stage-3 changes force rework and early products may not interoperate. Freeze dates (especially the ASN.1 freeze) mark when the specification is stable enough to implement without such risk.

13 Further reading

TR 21.900 [7] describes 3GPP working methods, version numbering and CR categories; TR 21.905 [8] is the vocabulary. The ITU-R documents on IMT-2020 and IMT-2030 [11]–[13] show how 3GPP's work connects to global requirements. Dahlman et al. [9] describe the standardisation history of NR.

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